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Separation of Ultra-High-Density Cell Suspension via Elasto-Inertial Microfluidics

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Separation of high-density suspension particles at high throughput is crucial for many chemical, biomedical, and environmental applications. In this study, elasto-inertial microfluidics is used to manipulate ultra-high-density cells to achieve stable equilibrium positions in microchannels, aided by the inherent viscoelasticity of high-density cell suspension. It is demonstrated that ultra-high-density Chinese hamster ovary cell suspension (>26 packed cell volume% (PCV%), >95 million cells mL^{-1}) can be focused at distinct lateral equilibrium positions under high-flow-rate conditions (up to 10 mL min^{-1}). The effect of flow rates, channel dimensions, and cell densities on this unique focusing behavior is studied. Cell clarification is further demonstrated using this phenomenon, from 29.7 PCV% (108.1 million cells mL^{-1}) to 8.3 PCV% (33.2 million cells mL^{-1}) with overall 72.1% reduction efficiency and 10 mL min^{-1} processing rate. This work explores an extreme case of elasto-inertial particle focusing where ultra-high-density culture suspension is efficiently manipulated at high throughput. This result opens up new opportunities for practical applications of high-particle-density suspension manipulation.

1. Introduction

Precise and efficient manipulation of high-density particle suspension is critical for applications in chemical and biomedical fields (e.g., biomanufacturing, medicine, diagnostics). Although membrane filtration is commonly used for the separation of high-density particle suspension, it has many limitations, such as membrane fouling and clogging.[1–4] These challenges become more severe for higher density suspensions such as blood and cell culture in bioprocessing, highlighting the need to develop membrane-less particle manipulation techniques with simplicity, good cell viability, and high throughput.

Inertial microfluidics has been a powerful approach to control particle motion in Newtonian fluids at high throughput.[5,6] Recently, the combination of this fluid inertia and fluid elasticity has been utilized to manipulate particles further in microchannels (i.e., elasto-inertial microfluidics).[7–11] This led to many exciting applications in particle focusing[12–23] and separation.[14–16] However, in most prior experiments, viscoelasticity of the cell solution was artificially generated by polymeric additives.[12–18,20–23] Yet, many cell suspension samples of practical implication have high cell density and thus exhibit intrinsic viscoelasticity (e.g., blood[24–27] and culture suspension[28–30]). Therefore, direct elasto-inertial manipulation may be possible. Nevertheless, the importance of cell density in elasto-inertial particle focusing has not been fully understood nor investigated. Prior studies in cell focusing and separation in elasto-inertial flows were limited to relatively low cell densities (<3 packed cell volume% (PCV%)).[12–16,21,22]

In this study, we demonstrated the first example of elasto-inertial particle focusing at ultra-high cell density in microchannels, without adding any polymeric additives. Using mammalian cell culture suspension (Chinese hamster ovary (CHO) cells), we demonstrated that ultra-high-density suspension cells provide inherent viscoelasticity of the fluid, which allows unique focusing behavior of the cells specific to high cell density suspensions. Using this unique elasto-inertial cell focusing, efficient cell clarification by the spiral microfluidic devices has been demonstrated, with significant density reduction (72.1%) at ultra-high-density CHO cell suspension (29.7 PCV%, 108.1 million cells mL^{-1}). This work clearly shows that high-density cell suspension can be effectively manipulated in microchannels at high throughput utilizing inherent viscoelasticity, enabling various practical, and industrial applications.

2. Theorectical Background

2.1. Elasto-Inertial Focusing

Combination of inertial and elastic focusing at finite fluid inertia and elasticity regimes enables unique rigid particle or cell focusing at high throughput (Elasto-inertial focusing).[7–11]

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Two non-dimensional numbers, the channel Reynolds number (Re) and Weissenberg number (Wi), measure the inertial and elastic effect of the fluid, respectively: $Re = \frac{\rho_f V D_h}{\eta}$ where ρ_f is the fluid density; V is the average fluid velocity; D_h is the hydraulic diameter of the channel; η is the dynamic viscosity of the fluid. $Wi = \lambda \dot{\gamma} = \lambda 2V/D_h$ where λ is the relaxation time of the fluid; $\dot{\gamma}$ is the characteristic shear rate. The elasto-inertial effect is often expressed using another non-dimensional number, Elasticity number (El), as follows: $El = \frac{Wi}{Re} = \frac{2\lambda\eta}{\rho_f D_h^2}$

Viscoelasticity, one of the unique properties of non-Newtonian fluid,[31] is a key element of elasto-inertial focusing. It causes elastic lift force on particles to drive them laterally to the equilibrium positions. Elastic lift force (F_{eL}) arising from this viscoelasticity of the fluid is expressed as follows:

$$F_{eL} = -2 f_{eL} a^3 \eta_p \lambda \nabla \dot{\gamma}^2 \tag{1}$$

where f_{eL} is the elastic lift coefficient; a is the spherical diameter of the particle; η_p is the polymeric contribution to the solution viscosity (η). The elastic lift force is applied in the direction toward the region with a smaller shear rate, typically the center of the microfluidic channel.

In addition, the fluid with finite inertia in microchannels causes inertial lift forces on the cells to migrate them laterally toward several equilibrium positions along the channel.[5,6] This inertial migration arises from two dominant size-dependent hydrodynamic forces: shear-gradient lift force and wall-induced lift force. They result from the parabolic flow profile of pressure-driven flows and particle-wall interaction, respectively. The net lift force, F_L, is expressed as

$$F_L = \frac{\rho_f V^2 a^4}{D_h} f_L(Re, x) \tag{2}$$

where $f_L(Re,x)$ is the lift coefficient of F_L depending on channel Reynolds number (Re) and lateral particle position (x) in the channel.

In addition to elastic and lift forces, the curved microchannels such as spirals induce lateral secondary-flow (Dean flow) drag force by generating two counter-rotating vortexes in the cross-section so that cell focusing along the channel can be further controlled laterally.[32,33] This drag force is expressed as:

$$F_D = 3\pi \eta_0 U_{De} a \tag{3}$$

where $U_{De} = 1.84 \times 10^{-4} De^{1.63} (m\ s^{-1})$ is the average transverse Dean velocity; $De = Re\sqrt{\frac{D_h}{2R}}$ is the non-dimensional number representing the strength of the Dean flow.

3. Results

3.1. Focusing of CHO Cells in High-Density Cultures

An 8-loop spiral microfluidic device with a trapezoidal cross-section (inner depth: 179 µm, outer depth: 110 µm, width: 1500 µm) was used to examine the focusing behavior of CHO cells at three different cell densities (10.1, 18.9, and 26.1 PCV% corresponding to 39.5, 70.3, and 95.5 million cells mL^{-1} (MC mL^{-1})) (**Figure 1**). The device had one inlet and two outlets. The concentrated CHO cells (average size: 17 µm) in culture medium at the desired density flowed into the inlet of the device at 6 mL min^{-1}. The focusing behavior of the CHO cells was observed near the outlets (**Figure 2A**). The shift of focusing position was observed at all three different PCV%, creating a density gradient laterally. At the lowest PCV% (10.1 PCV%, 39.5 MC mL^{-1}), CHO cells occupied the space near the inner wall while making a cell-free region near the outer wall. By contrast, at 18.9 PCV% (70.3 MC mL^{-1}), the CHO cells were found to be more focused near the outer wall, although the cells filled the entire channel in the lateral direction (no cell-free region) which was previously reported by Goh et al.[34,35] The peak intensity position near the outer wall was observed (Figure 2B). As the density was further increased to 26.1 PCV% (95.5 MC mL^{-1}), more focusing of the cells near the centerline of the channel was observed. Although the cell-free region was still not formed, the cell density near both sides was lower than that near the centerline, producing a lateral cell density gradient (Figure 2A,B).

Video S1, Supporting Information shows the patterns of focusing near the centerline at higher cell density (35.1 PCV%, 127 MC mL^{-1}). Cells are not focused on specific line and focusing streak has wide width due to increased cell-to-cell interaction at high cell densities. Narrow cell focusing can be observed at low cell density (<5 PCV%, Figure S1, Supporting Information). High flow rate (up to 10 mL min^{-1} in this study) did not affect cell viability (Figure S2, Supporting Information).

3.2. Focusing of CHO Cells in Cultures with Different Viscoelasticity

To understand whether the patterns of CHO cell focusing by cell density are likely attributable to the differences in their viscoelasticity, we examined the patterns in a typical viscoelastic fluid with different levels of viscoelasticity. Using the same microfluidic channel, poly-ethylene oxide (PEO) was added to the culture medium at different concentrations (0.1 wt% and 1 wt%) to create different levels of viscoelasticity. To minimize the effect of interaction between cells, input cell density of 0.5 MC mL^{-1} (<0.1 PCV%) was used. At the same flow rate conditions (6 mL min^{-1} input flow rate), similar patterns of position shift were clearly observed (Figure 2C). The CHO cells in the culture medium without PEO (i.e., lowest viscoelasticity) were focused near the inner wall of the channel, which is similar to the patterns observed for the lowest cell-density culture. By contrast, the cells in the medium containing 0.1 wt% PEO were shifted to near the outer wall of the channel. Interestingly, higher PEO concentration (1 wt%) caused the cells to be focused near the centerline of the channel, exhibiting the typical focusing behavior of the cells in the highly viscoelastic fluid. Figure 2D describes the distinct cell focusing behavior in culture suspensions with different viscoelasticity.

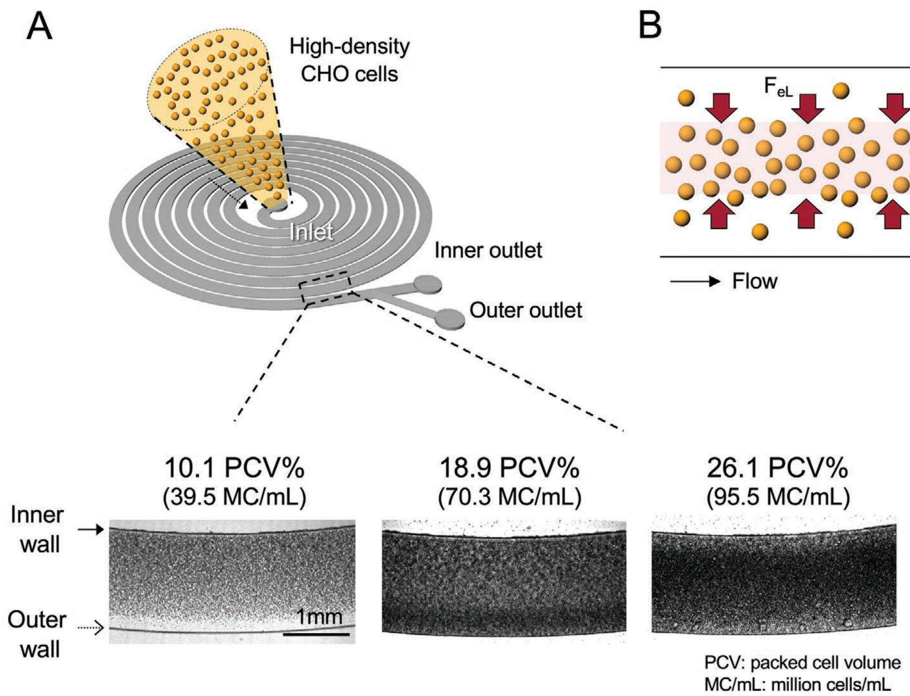

Figure 1. Focusing of CHO cells in high-density cultures in the spiral microchannel. A) Unique cell focusing in the spiral microfluidic channel at different cell densities. The input flow rate was 6 mL min^{-1}. B) Contribution of the elastic lift force (F_{eL}) arising from viscoelasticity of high-density CHO cell suspension to cell focusing in the channel. Inertial lift (F_L) and Dean drag (F_D) forces are not shown in the figure.

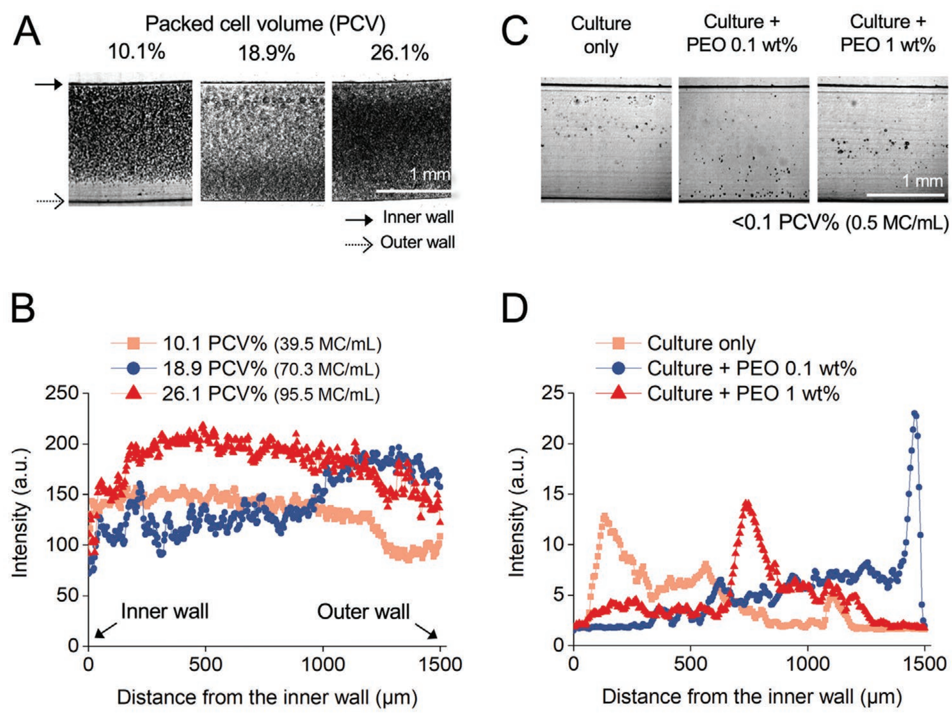

Figure 2. Analogy of cell focusing between high-density culture suspension and PEO-added (viscoelastic) culture suspension in the spiral microchannel. A) Cell focusing shift according to density (PCV%) in high-density culture suspension. B) Lateral cell focusing profile from the top view for high-density culture suspension. C) Cell focusing shift according to PEO concentration (different levels of viscoelasticity) in culture suspension. The CHO cell density was 0.5 million cells mL^{-1} (<0.1 PCV%). D) Lateral cell focusing profile from the top view for PEO-added culture suspension.

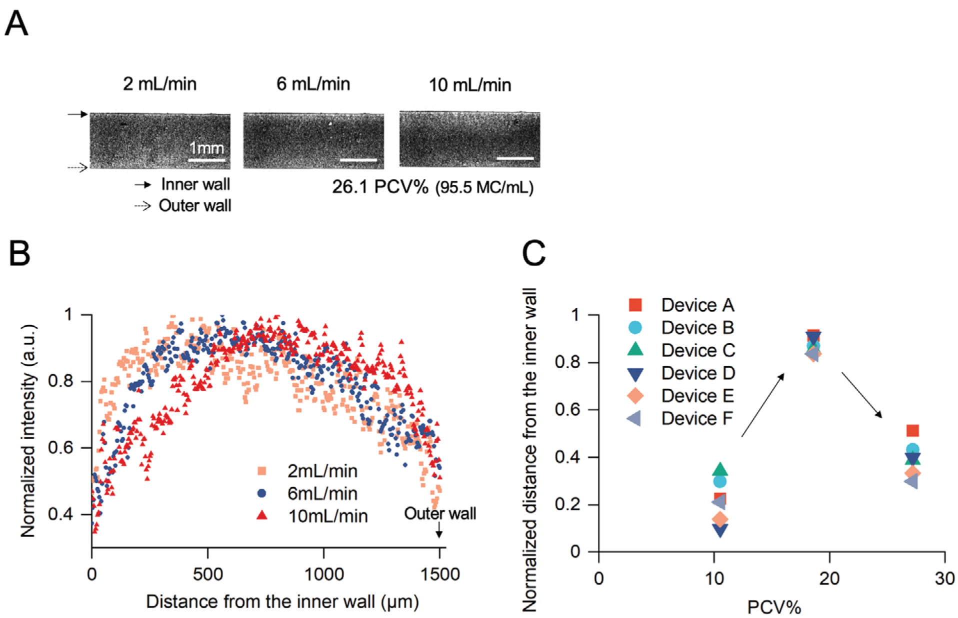

Figure 3. Focusing of CHO cells at varying input flow rates and cross-sectional channel geometries. A) Cell focusing shift at three different input flow rates (2, 6, and 10 mL min^{-1}) in the 179 µm/110 µm/1500 µm (inner depth/outer depth/width) spiral microchannel. The high-density (PCV 26.1%, 95.5 MC mL^{-1}) culture suspension was used. B) Lateral cell focusing profile from the top view according to different input flow rates. C) Position of the focusing peak at different input cell densities in six microchannels. The dimensions are described in Table 1. The actual cell focusing images are in Figure S4, Supporting Information.

3.3. Focusing of CHO Cells at Varying Input Flow Rates and Channel Geometries

Next, we investigated the effect of different flow rates and channel cross-sectional dimensions on focusing of high-density CHO cells. First, three different flow rates (2, 6, and 10 mL min^{-1}) were tested using the high-cell-density (26.1 PCV%, 95.5 MC mL^{-1}) culture suspension (**Figure 3A**). At this high cell density, all three flow rates formed the focused cell band away from the outer wall of the channel, focusing the cells near the centerline. As the input flow rate increased, the peak of the focused cell band moved toward the centerline of the channel (peak position of 392, 608, and 794 µm at 2, 6, and 10 mL min^{-1}, respectively) (Figure 3B), suggesting that more elastic lift force was generated due to increased shear rate ($F_{eL} \approx \dot{\gamma}^2$). At 6 and 10 mL min^{-1}, reduced cell density near both wall sides of the channel was clearly observed. Figure S3, Supporting Information describes cell focusing shift at lower density (10 and 19 PCV%).

Table 1. Spiral microchannels with different cross-sectional dimensions and flow conditions used for Figure 3C.

Device	Inner depth [µm]	Outer depth [µm]	Width [µm]	Cross-sectional area [×10^6 µm^2]	Flow rate [mL min^{-1}]
A	195	142	1500	2.53	8
B	179	110	1500	2.17	8
C	191	148	1000	1.70	4
D	183	139	1000	1.61	4
E	211	123	1000	1.67	4
F	237	118	1000	1.78	4

To examine whether the shifting of focused cell band is specific to a given microchannel, we varied the cross-sectional dimensions to the inner depth of 179–237 µm, outer depth of 110–148 µm, and width of 1000–1500 µm (**Table 1**). Regardless of the dimensions, a similar shift of the focusing position was observed with increased culture cell density (10.5 to 27.2 PCV%, 40.9 to 99.3 MC mL^{-1}) (Figure 3C and Figure S4, Supporting Information). The shifting of the focused cell band was from the inner wall to the outer wall, finally toward the centerline of the channel. It is also noted that the centerline focusing of the CHO cells in the 1000 µm channels (Devices C, D, E, and F) was not dependent on the slope of the cross-section.

3.4. Separation of CHO Cells Using Fluid Split at the Bifurcation

As the focusing of CHO cells in spiral microchannels creates low- and high-density streams, separation of cells (i.e., cell clarification) is possible by splitting these two streams through the bifurcated outlets. For example, in higher cell density culture, fewer cells flow near the inner wall, and more cells flow near the centerline or outer wall (Figure 2A). Accordingly, the low-cell-density stream can be collected from the inner outlet, whereas the high-cell-density streams can be collected from the outer outlet. The location of the boundary between these two outlet streams (i.e., fluid split) can be modulated by adjusting the fluidic resistance of the outlets (**Figure 4A**). The boundary becomes closer to the inner wall with increasing fluidic resistance and decreasing proportion of the inner outlet flow.

We investigated the magnitude of cell density reduction in the inner outlet of the spiral microchannel by varying levels of fluid split (IO%:OO% = 11%:89%, 24%:76%, and

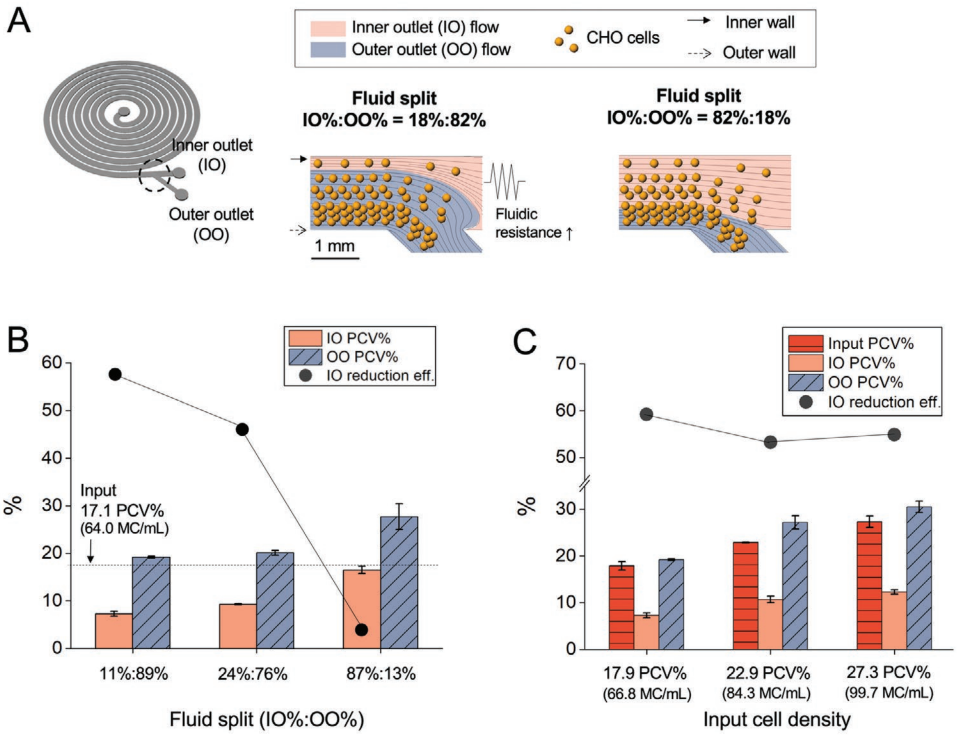

Figure 4. Separation of CHO cells using fluid split at the bifurcation. IO and OO are the inner outlet and outer outlet, respectively. A) Modulation of output streamline boundary by adjusting fluid resistance. The streamlines were obtained by computational flow simulation of two different fluid split cases (IO flow%:OO flow% = 18%:82% and 82%:18%). The CHO cell density gradient (higher density near the outer wall) arising from cell focusing at high cell density (e.g., 17.1 PCV%) was represented. B) Density reduction at different fluid split values. Input cell culture at 17.1 PCV% was flowed into the device at 10 mL min^{-1}. 10 mL min^{-1} was chosen to improve throughput. Error bars, standard deviation (n = 3). C) Density reduction at three different input cell densities (17.9, 22.9, and 27.3 PCV%) with a fixed fluid split (IO%:OO% = 11%:89%). The input flow rate was 10 mL min^{-1}. Error bars, standard deviation (n = 3).

87%:13%) (Figure 4B). The levels of fluid split were controlled by increasing fluidic resistance with attaching long and small-diameter (≈500 µm inner diameter) tubing to the outlet. Split values were confirmed by measuring the collected sample volumes. We used an input cell density of 17.1 PCV% (64.0 MC mL^{-1}) where the peak of the focused cell band is formed near the outer wall. At a fixed flow rate of 10 mL min^{-1}, density of the CHO cell culture collected in the inner outlet was 7.3 PCV% (29.7 MC mL^{-1}), 9.3 PCV% (36.7 MC mL^{-1}), 16.5 PCV% (61.9 MC mL^{-1}), respectively, for each fluid split condition. We calculated reduction efficiency as the ratio of the inner outlet density to the input cell density (see Experimental Section). For each fluid split, reduction efficiency in the inner outlet was 57.3%, 45.6%, and 3.5%, respectively. The smallest proportion of inner outlet flow (i.e., 11%:89% fluid split) had the lowest cell density with the highest reduction efficiency.

We further examined cell density reduction in three higher cell densities (17.9, 22.9, and 27.3 PCV% (66.8, 84.3, and 99.7 MC mL^{-1})) with a fixed fluid split of 11%:89% (IO%:OO%) (Figure 4C). Again, cell density reduction through the inner outlet was observed for all three densities at 10 mL min^{-1} input flow rate: (1) 17.9 to 7.3 PCV%, (2) 22.9 to 10.7 PCV%, and (3) 27.3 to 12.3 PCV%. The reduction efficiency was 59.2%, 53.3%, and 54.9%, respectively. Reduction efficiency did not decrease linearly with increased cell density. In all three cultures, cell density in outer outlets was higher than their input density, showing lateral suspension fractionation from cell focusing.

3.5. Separation of CHO Cells in Ultra-High-Density Suspension Culture Using Cascaded Configuration

Cell reduction efficiency at high-density suspension can be further improved by using a cascaded configuration. **Figure 5A** shows the schematic of this cascaded configuration using two spiral microchannels. The first device reduces the cell density for the second device. Overall reduction efficiency can be further enhanced after the second cell reduction from the second device. As shown in Figure 4, the majority of the focused cells at high cell density (e.g., >17 PCV%, >63.7 MC mL^{-1}) can be collected in the outer outlet of the device. The remaining cells are collected in the inner outlet. These cells can flow again into the second device for further density reduction. The actual two devices in series are shown in Figure 5B. The inner outlet of the first device was connected to the inlet of the second device through silicone tubing. Considering fluid split from the first device, the spiral channel with smaller cross-sectional dimensions was used as the second device for effective cell focusing at lower input flow rates (Figure S5, Supporting Information). The input flow rates for the first and second devices were 10 and 1.5 mL min^{-1}, respectively. The fluid split values for

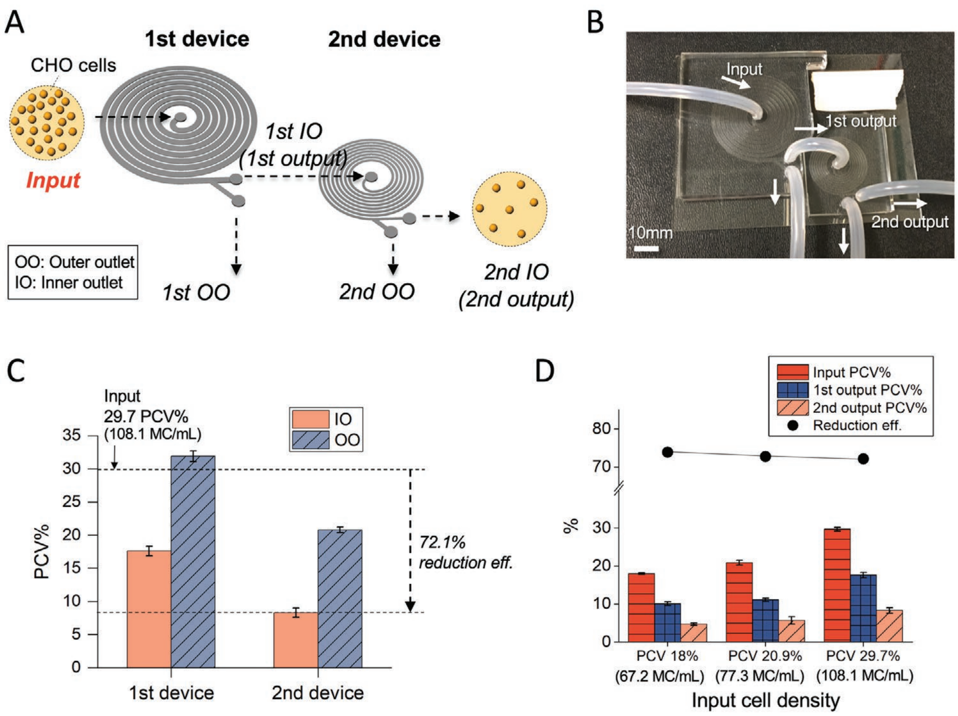

Figure 5. Separation of CHO cells in ultra-high-density suspension culture using cascaded configuration. A) Schematic of two spiral devices in series to reduce cell density of ultra-high cell density suspension. The dimensions for the first and second devices are 179 µm/110 µm/1500 µm and 84 µm/119 µm/600 µm, respectively (inner depth/outer depth/width). B) The actual photo of two spiral devices in cascaded configuration. C) Density of the samples collected from the two spirals in series. The density of the first IO (first output) sample was estimated by the fluid split and outlet densities of the second device. The input cell density of the first device was 29.7 PCV% (108.1 MC mL^{-1}). The input flow rates for the first and second devices were 10 and 1.5 mL min^{-1}, respectively. The fluid split values for the first and second devices were IO%:OO% = 15%:85% and 26%:74%, respectively. Error bars, standard deviation (n = 3). D) Density reduction at three different input cell densities using the cascaded configuration. The first output corresponds to the culture suspension flowing through the inner outlet (IO) of the first device (i.e., input for the second device). The second output is the culture suspension collected from the inner outlet of the second device. Error bars, standard deviation (n = 3).

the first and second devices were IO%:OO% = 15%:85% and 26%:74%, respectively. The flow rate of the inner outlet of the second device (final output) was 0.39 mL min^{-1} (=562 mL day^{-1}).

With the flow conditions described above, the CHO cell suspension at 29.7 PCV% (108.1 MC mL^{-1}) flowed into the cascaded devices. The final output (inner outlet of the second device) had the lowest density (8.3 PCV%, 33.2 MC mL^{-1}), achieving 72.1% overall reduction efficiency (Figure 5C). Compared with the earlier result without cascaded configuration (54.9% efficiency at 27.3 PCV%, 99.7 MC mL^{-1}), the reduction efficiency increased by 17.2 percentage points even with the increased input cell density from 27.3 to 29.7 PCV%. Using the fluid split and measured PCV%, the input density of the second spiral device was estimated to be 17.6 PCV% (65.8 MC mL^{-1}). Therefore, the reduction efficiencies of the first and second devices were 40.7% and 52.8% at 29.7 PCV% and 17.6 PCV%, respectively. As the cell density reduction depends on the level of fluid split (Figure 4B), the reduction efficiency at this high cell density could be further improved by using lower flow rate for the inner outlet of the second device.

Figure 5D summarizes the cell reduction performance of the cascaded spiral configuration at three different input cell densities (18.0, 20.9, and 29.7 PCV% corresponding to 67.2, 77.3, and 108.1 MC mL^{-1}, respectively). The final cell density and overall reduction efficiency are as follows: 1) For 18.0 PCV%, 4.7 PCV% and 73.9%, 2) For 20.9 PCV%, 5.7 PCV% and 72.7%, and 3) For 29.7 PCV%, 8.3 PCV% and 72.1%.

4. Discussion

This work demonstrates the focusing of ultra-high-density suspension cells in the elasto-inertial fluid flow in spiral microchannels. We applied this phenomenon to high-cell-density CHO cell separation, which is an important task in the biopharmaceutical industry. This focusing was achieved using the intrinsic viscoelastic property of high-density cell suspension without synthetic or biological elasticity enhancers (e.g., high-molecular-weight polymers or hyaluronic acid). This work is one of the few examples of high-throughput elasto-inertial focusing, where the input flow rate is on the order of mL min^{-1}.[13] For practical applications, we demonstrated significant density reduction of high-density CHO cell suspension using fluid split control and cascaded configuration.

In elasto-inertial fluid flows of the spiral microchannels, particles or cells are focused and separated by their size based on a combination of elastic lift, inertial lift, and Dean drag forces.[18,36] Based on the balance between these three forces, the particles or cells occupy distinct equilibrium positions along the spiral channel. The typical position could be near the inner wall of the spiral channel when elastic force is negligible (i.e., no viscoelastic fluid flow).[33] However, additional fluid elasticity could also vary the equilibrium positions.[18,36] Xiang et al. studied the particle migration mechanism in spiral microchannels and demonstrated that additional fluid elasticity can shift the particle focusing position from the inner wall to the outer wall in spiral microchannels.[18] Furthermore, it was confirmed that significant fluid elasticity arising from high concentration of a synthetic elasticity enhancer (Polyvinylpyrrolidone, or PVP) can shift the particles from the outer wall to the centerline of the spiral microchannels.[36]

Considering these earlier works[18,33,36] and analogy of cell focusing behavior between high-density culture suspension and PEO-added culture suspension (Figure 2), the focusing shift of the high-density suspension cells is likely due to increased elastic lift force arising from viscoelasticity of the high-density suspension, which is further increased at higher cell densities. Viscoelastic behavior of high-density mammalian cell suspension, such as blood and concentrated CHO cell suspension, has been well documented in prior studies.[27,29,30,37] Cell elasticity and cell-to-cell adhesion energy are two major mechanisms.[30,37] The membrane of the cells can restore the integrity of the cells reversibly (cell elasticity). In addition, elastic coupling between cells occurs due to increased cell interactions at high density (cell-to-cell adhesion energy). In a prior study, elastic plateau modulus of the concentrated CHO cell suspension was modeled by considering CHO cell's effective elastic modulus and adhesion energy.[29] In this study, we observed cell clusters at high-density CHO cell suspension because of close packing (Figure S6, Supporting Information). Higher cell density increases both relaxation time (λ) of the suspension and fluid viscosity (η_p),[29] thereby leading to increased elastic lift force (F_{eL}) on the cells in microchannels. As the elastic force directs toward the region of smaller shear rate, the cells are focused to the centerline of the channel at high cell density.[11]

We observed two different cell focusing positions (near inner and outer wall) at lower cell densities before centerline focusing at higher cell densities. Inner wall focusing at low cell density (e.g., <10.1 PCV%) occurs presumably because the inertial lift and Dean drag forces are dominant over the elastic lift force. By contrast, as the cell density is increased (e.g., 18.9 PCV%), the focused cell band is formed near the outer wall of the channel where elastic lift force arising from high cell density suspension is increased. Additional viscoelasticity of the fluid renders hydrodynamic radius of particles larger, promoting cells to focus toward the outer side of the trapezoidal spiral microfluidic channel.[38] Finally, the shifting of the focused cell band toward the centerline of the channel (e.g., 26.1 PCV%) could be due to further increased elastic lift force dominating over other two forces (Dean drag and inertial lift forces), resulting from the viscoelasticity of the ultra-high-density cell suspension. The competition between the inertial lift and elastic lift force can be evaluated using the Elasticity number ($El = Wi/Re = 2\lambda\eta/\rho_f D_h^2$); At 26.1 PCV%, El was estimated to be at least 14 (using measured λ of >100 ms and η of 5 mPa·s; λ of >400 ms at 20 PCV% was reported previously[29]), showing dominant elastic lift force over inertial force.

Using continuous elasto-inertial fluid flows in microchannels can create new opportunities for efficient and high throughput manipulation of high-density cell suspensions without the use of membrane filters/centrifuge or external elasticity enhancers. Although centrifugation is the most commonly used and efficient method for liquid-solid separation, continuous-flow operation and separation of different cell populations are challenging. On the contrary, current microfluidic device is suitable for the applications requiring continuous-flow cell separation (e.g., continuous bioprocessing; see below). In addition, the device can separate single cells and large cell clusters effectively at high cell densities (Figure S6, Supporting Information). In typical elasto-inertial microfluidics studies, synthetic polymers are added to the Newtonian fluid (e.g., phosphate buffer or water) to induce fluid elasticity.[7–11] However, this addition can be considered as chemical contamination;[11] it could affect cell viability and interfere with downstream bioassays.[11] In this context, we evaluated the potential of using cell density as a natural elasticity enhancer; many cell suspensions in real-world applications have high density and thus exhibit intrinsic viscoelasticity (e.g., blood[24–27] and culture suspension[28–30]). While there were a few studies regarding the focusing or separation of bioparticles at high PCV% (e.g., whole blood) in viscoelastic fluids,[39,40] their bioparticle manipulation was performed with the addition of fluid elasticity enhancers (e.g., PEO).[39–42] In fact, most earlier experimental studies of elasto-inertial flows were limited to low cell densities (<3 PCV%).[12–16,21,22]

As a way to further improve elasto-inertial cell focusing in high-density suspension in spiral channels, the experiments with fluid-particle simulation[43] and modification of channel geometries (e.g., cross-sectional shape[22,23]) could be contemplated. Interestingly, a recent simulation study in moderate fluid inertia (Re = 37.8) and small elasticity (Wi up to 2) fluid flow showed enhanced focusing of deformable cell suspension at high PCV% (e.g., 20 PCV%) with increased fluid elasticity (Wi).[43] Moreover, the effect of channel cross-sectional shapes on cell focusing was studied.[22,23] For example, noting inherent viscoelasticity of the blood at high PCV% (30 PCV%), Chen et al. recently studied the margination of stiffened red blood cells in microchannels with various cross-sectional shapes.[24,25]

Our findings have many immediate applications, including cell retention from continuous perfusion bioreactors.[44] Continuous solid-liquid separation is routinely performed by hollow fiber membrane-based cell retention devices during perfusion culture in continuous biomanufacturing;[44–46] the cell retention device retains cells in the bioreactor and separates out cell-less liquid containing toxic metabolites and therapeutic proteins from the bioreactor.[44,47,48] We previously developed a new membrane-less cell retention device based on inertial cell sorting.[49] Nevertheless, the demonstrated cell retention capacity was limited to ≈40 million cells mL^{-1},[49,50] which is lower than the capacity of membrane-based cell retention (50–100 million cells mL^{-1}). Our current study shows the potential of enhancing cell retention capacity by microfluidic devices in the elasto-inertial fluid regime, as demonstrated by Figure 5. Compared with other membrane-less cell manipulation techniques such as acoustic wave separators, centrifuges, hydrocyclones, and gravity settlers,[51,52] (Table S1, Supporting Information), the microfluidic cell retention device based on elasto-inertial cell focusing has many benefits. With its enhanced cell density capacity demonstrated in this study, the membrane-less microfluidic device can perform cell retention with advantages including removal of small dead cells/debris and low capital/operational expenses,[49,50,53–55] which could lead to efficient and reliable high-density perfusion culture at various scales.

There are several other potential applications of the phenomenon observed in this work, including manipulation of bio-nanoparticles[8,56] and focusing/separation of other high-density suspension bioparticles (e.g., microalgae culture and blood). Researchers recently demonstrated the manipulation of sub-micron particles[41,57–60] and DNAs[57,59] as well as cellular vesicles[41,57,58,60] such as exosomes in viscoelastic flows. Considering this demonstration and using innate viscoelasticity of high-density cell suspension, desired or undesired bio-nanoparticles such as extracellular vesicles from culture suspension could be sorted out (Figure S7, Supporting Information). Also, efficient and high-throughput microalgae biomass recovery,[61,62] blood fractionation,[63] and plasma extraction[63] could be potential future applications.

5. Conclusion

In summary, this work investigated the focusing of suspension cells at ultra-high density in microchannels, which has many potential applications yet has been unexplored in elasto-inertial microfluidics. We showed that high cell density in suspension culture could function as a natural elasticity enhancer and enable precise and high-throughput cell manipulation in the elasto-inertial fluid flow. Beyond synthetic elasticity enhancers used in conventional viscoelastic microfluidics, we expect that natural viscoelasticity arising from high cell density or various body fluids can find many meaningful applications in the future.

6. Experimental Section

CHO Cell Culture and Culture Sample Preparation: CHO-S cells (FreeStyle CHO-S Cells, R80007, Thermo Fisher Scientific, USA) were grown in glass spinner flasks (4500-500, Corning, USA) in a 5% CO_2 incubator. The cells were seeded at 0.3 million cells mL^{-1} to a 250–310 mL culture medium (FreeStyle CHO Expression Medium, 12651022, Thermo Fisher Scientific, USA) and maintained for up to 5–7 days. No anti-clumping agents or surfactants were used. The culture parameters such as density, viability, nutrients/metabolites concentration, aggregate formation, pH, and ions were regularly monitored using an automated cell culture analyzer (FLEX2, NovaBiomedical, USA). In order to prepare high-density cell culture, the cell culture was centrifuged (200 g for 3 min), and cell-free solution was removed. The PCV% was measured as below to measure high cell density (>30 million cells mL^{-1}) accurately in the presence of cell clusters.

$$\text{Packed cell volume}(\text{PCV\%}) = \frac{\text{volume of packed cells}(\text{mL})}{\text{total volume of culture suspension}(\text{mL})} \times 100 \quad (4)$$

This PCV% value was also converted to million cells mL^{-1} (MC mL^{-1}) based on calibration by the automated cell culture analyzer (FLEX2, NovaBiomedical, USA) (Figure S8, Supporting Information). The prepared high-density culture suspension was transferred into either 5 mL macrotubes (for large sample volume of >1 mL; 470225-006, VWR, USA) or hematocrit tubes (for small sample volume of <1 mL; 10007500-C/5, Drummond, USA). Subsequently, the tubes containing culture suspension were centrifuged at 1800 g for 5 min to settle cells.

PEO Sample Preparation: To test the focusing of CHO cells in polymer-based viscoelastic fluids, PEO was used (372781-5G, viscosity average molecular weight of 1,000 kDa, MilliporeSigma, USA). PEO-added cell culture solution was prepared at two different wt% (0.1 and 1 wt%) concentrations in culture medium (FreeStyle CHO Expression Medium, Thermo Fisher Scientific, USA). Based on an earlier study, the relaxation time (λ) of the PEO-added (1 wt%) culture was estimated to be 1.58 ms.[21] The original cell culture without PEO at 0.5 million cells mL^{-1} was centrifuged for 3 min at 200 g, and cell-free solution was then removed. Subsequently, the PEO containing cell-free medium was transferred to make viscoelastic cell culture suspension at 0.5 million cells mL^{-1}.

Microfluidic Device Design and Fabrication: The molds for microfluidic devices were designed using a 3D modeler (Rhinoceros, Robert McNeel & Associates, USA). One of the devices had 179 µm inner depth, 110 µm outer depth, 1500 µm width, and 8 loops. The spiral inner and outer diameters were 10 and 41 mm, respectively. To prevent cell clogging at the outlet channels, the channels were designed such that both outlet channels had similar or higher cross-sectional area than the main spiral channel (Figure S9, Supporting Information). No clogging was observed in the channel when high-density CHO cells were processed. The dimensions of other spiral devices are described in Table 1. The molds were fabricated with aluminum (Whits Technologies, Singapore). In order to fabricate microfluidic devices, standard soft lithography using polydimethylsiloxane (PDMS, Sylgard 184, Dow Corning, USA) was used. The PDMS was poured onto the mold and then cured at 150 °C for 20 min. Subsequently, the micropatterned PDMS piece was removed from the mold, and input and output reservoirs were made using punches (Integra LifeSciences, USA). It was then bonded to the PDMS-coated (500 µm thick) glass substrate using oxygen plasma treatment (FEMTO SCIENCE, Korea). The prepared microfluidic device was placed on a hotplate at 95 °C overnight for complete bonding. For cascaded configuration, two different spiral devices were used in series: 179 µm/110 µm/1500 µm (first device) and 84 µm/119 µm/600 µm (second device) (inner depth/outer depth/width). The inner outlet of the first device was connected to the input of the second device. The culture sample collected from the inner outlet of the second device was regarded as a final output.

Observation of Cell Focusing and Characterization of Microfluidic Device Performance: In order to deliver the culture suspension to the device, silicone tubings (EW-96410-14, Cole-Parmer, USA) were inserted to each input and output reservoirs of the device. The culture suspension was then loaded to a syringe (Becton Dickinson, USA), and a syringe pump (PHD 2000, Harvard Apparatus, USA) delivered the culture suspension to the device at various flow rates (up to 10 mL min^{-1}). The focusing behavior of the cells was monitored under the inverted microscope (IX51, Olympus, Japan) using a high-speed imaging camera (Phantom v9.1, Vision Research, USA). Fluid simulation was performed using COMSOL Multiphysics 5.5 (COMSOL, Inc., USA). To characterize the density reduction quantitatively, the PCV (see above equation) of the input and output samples was measured after centrifuging the cells in either macrotubes or capillary tubes. To vary the fluid split (inner outlet flow% versus outer outlet flow%), the fluidic resistance of the outlet was modulated using a smaller-diameter tubing (EW-06420-02, Cole-Parmer, USA). By attaching this tubing and changing its length, the fluid split was controlled. The volume of the cell suspension collected from each outlet was measured to obtain the actual fluid split values. The cell reduction efficiency of the microfluidic device was obtained using the following equation:

$$\text{Cell reduction efficiency}(\%) = \left(1 - \frac{\text{Output cell density}}{\text{Input cell density}}\right) \times 100 \quad (5)$$

where the output cell density corresponds to the cell density of the outlet of the device which contains reduced cell density. Typically, this outlet was the inner outlet of the spiral device when the PCV% was high (e.g., >20 PCV%). This reduction efficiency was often used as cell retention efficiency (%) in the context of cell retention for perfusion culture.[64]

Measurement of the Viscoelasticity of High-Density CHO Cell Suspension: A rheometer (Discovery Hybrid Rheometer 2 from TA Instruments) was used to measure the viscoelasticity of the high-density (43 and 30 PCV%) CHO cell suspension. An oscillatory-frequency mode was used to measure viscous (loss) and elastic (storage) modulus of the complex viscosity. 20 mm parallel plate (Peltier plate, stainless steel, serial number: 113178) was chosen as a geometry, and the gap was set to 400 µm. Frequency was logarithmically swept from 0.05 Hz to 5.0 Hz (strain = 5%). Table S2, Supporting Information and Figure S10, Supporting Information summarize the measurement results.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

T.K. designed and fabricated spiral microchannels; T.K. and J.H. designed the experiments; T.K. and K.C. analyzed and maintained the CHO cell line; T.K. performed overall experiments; K.C. performed cell viability experiments; T.K., K.C., and J.H. analyzed and discussed the data; T.K., K.C., and J.H. wrote the main manuscript text. All authors reviewed the manuscript.

Data Availability Statement

Research data are not shared.

Keywords

cell focusing, cell retention, cell suspension, elasto-inertial microfluidics, high density

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